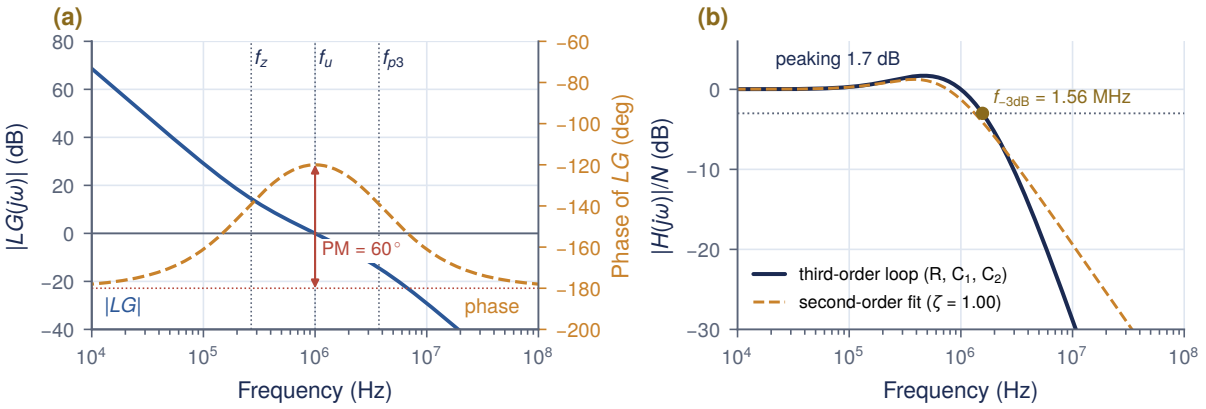
(a)
80
60
40
20
0
−20
−40
−60
−80
−100
−120
−140
−160
−180
−200
f_z
f_u
f_{p3}
PM = 60°
|LG|
phase
$|LG(j\omega)|$ (dB)
Phase of LG (deg)
10^4
10^5
10^6
10^7
10^8
Frequency (Hz)
(b)
peaking 1.7 dB
f_{-3dB} = 1.56 MHz
0
−10
−20
−30
third-order loop (R, C_1, C_2)
second-order fit (ζ = 1.00)
$|H(j\omega)|/N$ (dB)
10^4
10^5
10^6
10^7
10^8
Frequency (Hz)